

MANDALGOVI, Mongolia

WMO: 443410

Lat: 45.7431N

Lon: 106.2642E

Elev: 4570

StdP: 12.43

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-13.6	-26.2	1.5	-12.0	-22.5	1.9	-10.2	34.5	-2.1	30.4	4.5	8.2	340	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	88.7	60.6	85.3	59.1	82.1	57.9	65.2	80.4	62.8	77.9	60.9	74.9	12.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	92.7	70.6	57.9	84.9	67.9	55.7	78.5	66.6	32.8	80.3	30.8	77.7	29.3	75.5	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
34.2	29.4	26.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-22.1	94.3	4.9	3.3	-25.7	96.7	-28.6	98.6	-31.3	100.5	-34.9	102.9
			-22.3	66.6	4.8	4.2	-25.8	69.6	-28.6	72.0	-31.4	74.4	-34.9	77.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	1.7	9.1	24.6	40.7	53.1	64.6	69.8	65.9	54.5	37.9	18.6	5.3
	DBStd	25.33	8.16	9.77	10.67	9.48	9.19	7.21	6.00	6.50	8.55	9.17	10.46	8.19
	HDD50	6572	1497	1146	788	301	73	1	0	0	54	383	943	1386
	HDD65	10480	1962	1566	1253	729	380	93	21	68	325	839	1393	1851
	CDD50	1935	0	0	0	23	170	440	614	492	188	8	0	0
	CDD65	367	0	0	0	0	11	82	170	95	9	0	0	0
	CDH74	3728	0	0	0	4	204	914	1649	846	111	0	0	0
	CDH80	1164	0	0	0	1	38	278	590	244	13	0	0	0
Wind	WSAvg	11.0	9.3	10.0	12.2	13.9	14.0	12.4	10.7	10.0	10.3	10.5	9.3	8.9
Precipitation	PrecAvg	5.3	0.0	0.0	0.1	0.2	0.4	0.9	1.5	1.6	0.4	0.2	0.1	0.0
	PrecMax	9.8	0.2	0.4	0.3	0.7	1.9	3.7	4.1	4.5	2.2	1.7	0.3	0.2
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.0	0.1	0.1	0.2	0.4	0.7	0.9	1.1	0.5	0.3	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.3	38.1	58.5	72.2	84.4	91.5	95.0	90.9	82.0	67.1	48.9	30.6
		MCWB	21.7	28.6	39.9	47.1	55.4	58.9	63.5	59.6	55.6	45.9	35.6	23.7
	2%	DB	22.2	33.1	51.3	66.9	79.5	86.7	90.0	85.8	77.2	61.4	43.2	25.2
		MCWB	17.8	25.2	36.0	45.3	52.3	57.5	62.5	59.7	53.8	43.8	32.5	20.0
	5%	DB	18.1	28.8	46.2	62.3	75.3	83.0	86.3	82.3	73.1	57.0	38.3	21.3
		MCWB	14.6	22.5	33.6	43.1	50.2	56.9	60.7	57.9	52.4	41.3	29.6	17.2
	10%	DB	14.2	24.6	41.4	57.8	70.8	79.2	82.8	78.6	69.3	52.8	33.9	17.5
		MCWB	11.3	19.7	30.7	40.7	48.5	55.6	59.7	57.2	50.4	39.1	26.8	14.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	29.3	40.9	50.1	58.4	64.7	70.6	68.6	60.7	48.6	36.2	24.0
		MCDB	26.9	36.5	55.7	68.0	78.7	79.4	85.6	83.3	75.8	62.8	47.9	30.3
	2%	WB	17.9	25.6	36.9	46.4	54.6	61.7	66.3	63.5	56.6	44.8	32.9	20.3
		MCDB	21.8	32.5	49.8	63.5	72.9	78.0	82.2	77.4	71.5	60.0	42.9	24.9
	5%	WB	14.6	22.6	33.9	43.8	51.7	59.5	63.9	61.3	54.0	42.0	29.9	17.3
		MCDB	18.1	28.4	45.6	61.5	72.0	75.9	79.6	74.4	68.9	55.5	37.7	21.1
	10%	WB	11.3	19.7	30.9	41.1	49.3	57.5	62.0	59.6	51.8	39.5	27.1	14.2
		MCDB	14.1	24.4	41.3	56.6	69.0	74.4	76.9	71.7	67.1	51.9	33.7	17.4
Mean Daily Temperature Range	5% DB	MDBR	17.9	21.1	22.5	22.9	23.2	21.2	19.8	19.5	21.3	21.2	19.3	17.1
		MCDBR	21.8	25.7	27.3	28.8	28.7	25.9	24.6	24.1	25.5	25.8	23.3	20.5
		MCWBR	18.2	19.6	17.2	15.8	13.9	10.7	8.7	9.4	11.8	14.7	15.8	16.3
	5% WB	MCDBR	21.6	24.9	26.2	27.2	25.7	21.7	20.5	19.3	22.4	24.7	22.2	19.9
		MCWBR	18.1	19.2	17.1	15.6	13.9	10.5	9.6	9.5	11.5	14.7	15.4	16.1
Clear-Sky Solar Irradiance	taub	0.243	0.268	0.305	0.351	0.377	0.397	0.377	0.359	0.321	0.287	0.256	0.241	
	taud	2.379	2.326	2.268	2.187	2.156	2.193	2.311	2.337	2.391	2.419	2.411	2.381	
	Ebn at Noon	281	292	293	287	282	276	280	281	285	281	272	269	
	Edh at Noon	23	29	36	43	46	45	39	37	32	26	22	20	
All-Sky Solar Radiation	RadAvg	650	999	1514	1909	2226	2192	2086	1901	1596	1147	735	540	
	RadStd	53	74	52	95	109	111	87	80	54	43	48	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-4.43	N/A	-2.02	-2.14	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-3.37	N/A	-1.82	-1.53	N/A	N/A	N/A	N/A

Nomenclature: See separate page